Lessons 6–7. Vector Calculus I and II

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1 Vector Fields

1.1 Motivation

Before we dive straight into abstract definitions, let us pause and ask ourselves: why do we care about vector fields in the first place? The answer, as so often in mathematics, comes from physics — and from the very kind of differential equations we have been studying all along.

Recall the equation of a **simple harmonic oscillator**:

$$\frac{d^2x}{dt^2} + \omega^2 x = 0.$$

We can always rewrite a second-order ODE as a system of two first-order equations by introducing the velocity $v = \dot{x}$ as a second unknown:

$$\frac{dx}{dt} = v, \qquad \frac{dv}{dt} = -\omega^2 x.$$

The right-hand sides of these two equations together define a function that takes a point (x, v) in the plane and returns a *vector*:

$$\vec{F}(x, v) = \langle v,\ -\omega^2 x \rangle.$$

This is our first example of a **vector field**: at every point of the plane we have an arrow pointing in the direction the system will move next. The solutions of the ODE correspond exactly to the curves that are everywhere tangent to these arrows — so studying the geometry of the vector field is the same as studying the long-time behaviour of the differential equation. This is why vector fields are central to our subject.

1.2 Definition of Vector Fields

Let us now give the formal definition, which generalises the motivation above to an arbitrary number of dimensions.

Definition 1.1 (Vector Field)**.** A **vector field** on $\mathbb{R}^2$ (resp. $\mathbb{R}^3$) is a function $\vec{F}$ that assigns to each point (x, y) (resp. (x, y, z)) a two-dimensional (resp. three-dimensional) vector. We write

$$\vec{F}(x, y, z) = \langle P(x, y, z),\ Q(x, y, z),\ R(x, y, z) \rangle,$$

where P, Q, R are real-valued functions called the **component functions** of $\vec{F}$. Equivalently, using the standard basis vectors,

$$\vec{F}(x, y, z) = P(x, y, z)\,\hat{\mathbf{i}} + Q(x, y, z)\,\hat{\mathbf{j}} + R(x, y, z)\,\hat{\mathbf{k}}.$$

The vector field is said to be **continuous** if each of its component functions is continuous.

Geometrically, you should picture a vector field as a collection of arrows: at every point of space (or the plane) there is a little arrow indicating both a direction and a magnitude. Think of the wind velocity at every point of a weather map, or the gravitational pull exerted on a test mass at every point in space around the Earth — these are all vector fields that appear naturally in physics.

1.3 Gradient Fields

One of the most important ways in which a vector field can arise is as the *gradient* of a scalar function.

Definition 1.2 (Gradient Vector Field)**.** If f is a differentiable scalar function of x, y, and z, its **gradient** is the vector field

$$\nabla f(x, y, z) = \langle f_x,\ f_y,\ f_z \rangle = f_x\,\hat{\mathbf{i}} + f_y\,\hat{\mathbf{j}} + f_z\,\hat{\mathbf{k}}.$$

The gradient always points in the direction of the *steepest ascent* of f, and its magnitude equals the rate of change in that direction. Gradient fields are, in a very precise sense, the nicest vector fields that exist — and we will have a lot more to say about them shortly.

Example 1.1 (Computing a Gradient Field)**.** Find the gradient vector field of $f(x, y) = x^2 \sin(5y)$.

Solution. We simply differentiate:

$$\frac{\partial f}{\partial x} = 2x\sin(5y), \qquad \frac{\partial f}{\partial y} = 5x^2\cos(5y).$$

Therefore

$$\nabla f(x, y) = \langle 2x\sin(5y),\ 5x^2\cos(5y) \rangle.$$

Example 1.2 (Gradient Field and Level Curves)**.** Sketch the gradient vector field for $f(x, y) = x^2 + y^2$ together with several level curves.

Solution. The level curves are the sets $\{x^2 + y^2 = k\}$, which are concentric circles of radius $\sqrt{k}$ centred at the origin. The gradient is

$$\nabla f(x, y) = \langle 2x,\ 2y \rangle.$$

At any point (x, y), this vector points directly *away from* the origin — i.e. it is perpendicular to the circle through that point. This is a general rule: *gradient vectors are always perpendicular to the level curves (or level surfaces) of the function.* Moreover, the closer together the level curves are, the longer the gradient vectors, since the function is changing more rapidly.

1.4 Conservative Vector Fields

Definition 1.3 (Conservative Vector Field and Potential Function)**.** A vector field $\vec{F}$ is called **conservative** if there exists a scalar function f such that

$$\vec{F} = \nabla f.$$

The function f is called a **potential function** for $\vec{F}$.

Not every vector field is conservative. For example, $\vec{F} = \langle y, x \rangle$ is conservative with potential $f(x, y) = xy$. But $\vec{F} = \langle -y, x \rangle$ is not conservative — this is the field that "rotates" every vector 90 degrees counterclockwise, and no scalar function has this as its gradient.

The key practical question is: *how can we tell whether a given vector field is conservative, without actually finding a potential function?* The answer lies in checking a set of *cross-partial* conditions.

Theorem 1.1 (Cross-Partial Property of Conservative Vector Fields)**.** *Let $\vec{F} = \langle P, Q, R\rangle$ be a vector field whose component functions have continuous second-order mixed partial derivatives.*

- *In $\mathbb{R}^2$: if $\vec{F}(x,y) = \langle P(x,y), Q(x,y)\rangle$ is conservative, then*
$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$
- *In $\mathbb{R}^3$: if $\vec{F}(x,y,z) = \langle P, Q, R\rangle$ is conservative, then*
$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \qquad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}, \qquad \frac{\partial R}{\partial x} = \frac{\partial P}{\partial z}.$$

Proof. If $\vec{F} = \nabla f$, then $P = f_x$, $Q = f_y$, $R = f_z$. By the equality of mixed partials (Clairaut's theorem), $P_y = f_{xy} = f_{yx} = Q_x$, and similarly for the other pairs. □

Remark 1.1. This theorem gives a *necessary* condition for conservativity. On a simply connected domain (a notion we will define carefully in Section 3), the cross-partial condition is also *sufficient.* For now, if the cross-partials fail to be equal, we can immediately conclude the field is *not* conservative.

Example 1.3 (Testing Conservativity)**.** Determine whether $\vec{F} = \langle y^2,\ 2xy + e^{3z},\ 3ye^{3z}\rangle$ is conservative.

Solution. We identify $P = y^2$, $Q = 2xy + e^{3z}$, $R = 3ye^{3z}$ and check the three cross-partial conditions:

$$\begin{aligned} P_y &= 2y, & Q_x &= 2y, \quad \checkmark \\ Q_z &= 3e^{3z}, & R_y &= 3e^{3z}, \quad \checkmark \\ P_z &= 0, & R_x &= 0, \quad \checkmark \end{aligned}$$

All three conditions hold, so $\vec{F}$ *is* conservative.

Example 1.4 (Finding a Potential Function)**.** Show that $\vec{F} = \langle 2xy+z^3,\ x^2+2y,\ 3xz^2-5\rangle$ is conservative and find its potential function.

Solution.

Step 1: Verify conservativity. With $P = 2xy + z^3$, $Q = x^2 + 2y$, $R = 3xz^2 - 5$:

$$P_y = 2x = Q_x, \quad Q_z = 0 = R_y, \quad P_z = 3z^2 = R_x.$$

All conditions are satisfied, so $\vec{F}$ is conservative.

Step 2: Integrate. Since $f_x = P = 2xy + z^3$, integrate with respect to x:

$$f(x,y,z) = x^2y + xz^3 + g(y,z),$$

where g is an arbitrary function of y and z (the "constant of integration" in the x-variable).

Step 3: Use $f_y = Q$. Differentiating:

$$f_y = x^2 + g_y(y,z) = x^2 + 2y \implies g_y = 2y \implies g(y,z) = y^2 + h(z).$$

Step 4: Use $f_z = R$. We have $f = x^2y + xz^3 + y^2 + h(z)$, so:

$$f_z = 3xz^2 + h'(z) = 3xz^2 - 5 \implies h'(z) = -5 \implies h(z) = -5z + C.$$

Conclusion. The potential function is

$$f(x,y,z) = x^2y + xz^3 + y^2 - 5z + C.$$

2 Line Integrals

Now that we have a firm grasp of vector fields, we want to *integrate* them. The natural setting for this is integration along a *curve* in space, which leads to the concept of a line integral. There are two flavours: the **scalar line integral**, which integrates a scalar function along a curve, and the **vector line integral**, which integrates a vector field along a curve.

2.1 Scalar Line Integrals

Imagine walking along a curved path C in the plane (or in space) and at every point measuring the height of some function f. A scalar line integral adds up all these values, weighted by the arc length of the path. The construction mirrors that of an ordinary Riemann integral.

Theorem 2.1 (Scalar Line Integral). *Let C be a smooth curve parametrised by $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$, and let f be a continuous function whose domain contains C. Then the **scalar line integral** of f along C is*

$$\int_C f(x,y,z)\,ds = \int_a^b f\big(\vec{r}(t)\big)\,\|\vec{r}\,'(t)\|\,dt,$$

where $\|\vec{r}\,'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$ is the speed of the parametrisation.

The quantity $\|\vec{r}\,'(t)\|\,dt$ is just the infinitesimal arc length ds, so the formula is saying: "replace every point on the curve by $\vec{r}(t)$, and replace ds by the arc-length element of the parametrisation".

Example 2.1 (Scalar Line Integral along a Helix). Evaluate $\int_C (x + y^2)\,ds$, where C is the helix parametrised by $\vec{r}(t) = \langle \sin t, \cos t, t\rangle$, $0 \le t \le 2\pi$.

Solution.

Step 1: Compute the speed.

$$\vec{r}\,'(t) = \langle \cos t, -\sin t, 1\rangle, \qquad \|\vec{r}\,'(t)\| = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}.$$

Step 2: Substitute. On C we have $x = \sin t$ and $y = \cos t$, so $x + y^2 = \sin t + \cos^2 t$:

$$\int_C (x + y^2)\,ds = \int_0^{2\pi} (\sin t + \cos^2 t)\,\sqrt{2}\,dt.$$

Step 3: Integrate.

$$\sqrt{2}\int_0^{2\pi} \sin t\,dt + \sqrt{2}\int_0^{2\pi} \cos^2 t\,dt = \sqrt{2}\cdot 0 + \sqrt{2}\cdot \pi = \sqrt{2}\,\pi.$$

2.2 Vector Line Integrals: Work

The more physically important type of line integral is the *vector* (or *work*) line integral. If $\vec{F}$ is a force field and a particle travels along a curve C, the work done by the force is given by the following definition.

Definition 2.1 (Vector Line Integral)**.** Let $\vec{F} = \langle P, Q, R \rangle$ be a continuous vector field and let C be a smooth oriented curve with parametrisation $\vec{r}(t)$, $a \le t \le b$. The **vector line integral** (or **work integral**) of $\vec{F}$ along C is

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}\big(\vec{r}(t)\big) \cdot \vec{r}\,'(t)\, dt.$$

In terms of components this becomes

$$\int_C \vec{F} \cdot d\vec{r} = \int_C P\, dx + Q\, dy + R\, dz.$$

Geometrically, $\vec{F} \cdot d\vec{r} = \vec{F} \cdot \vec{T}\, ds$ where $\vec{T}$ is the unit tangent vector, so the integral measures how much $\vec{F}$ is aligned with the direction of travel along C.

Example 2.2 (Work Done by a Force Field)**.** How much work is required to move an object in the vector force field $\vec{F} = \langle yz, xy, xz \rangle$ along the path $\vec{r}(t) = \langle t^2, t, t^4 \rangle$, $0 \le t \le 1$?

Solution.

Step 1: Compute $\vec{r}\,'(t)$.

$$\vec{r}\,'(t) = \langle 2t, 1, 4t^3 \rangle.$$

Step 2: Evaluate $\vec{F}$ on the curve. Substituting $x = t^2$, $y = t$, $z = t^4$:

$$\vec{F}(\vec{r}(t)) = \langle t \cdot t^4,\ t^2 \cdot t,\ t^2 \cdot t^4 \rangle = \langle t^5, t^3, t^6 \rangle.$$

Step 3: Compute the dot product.

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t) = t^5 \cdot 2t + t^3 \cdot 1 + t^6 \cdot 4t^3 = 2t^6 + t^3 + 4t^9.$$

Step 4: Integrate.

$$W = \int_0^1 (2t^6 + t^3 + 4t^9)\, dt = \left[\frac{2t^7}{7} + \frac{t^4}{4} + \frac{4t^{10}}{10}\right]_0^1 = \frac{2}{7} + \frac{1}{4} + \frac{2}{5} = \frac{40 + 35 + 56}{140} = \frac{131}{140}.$$

2.3 Flux Across a Plane Curve

In two dimensions, there is another natural quantity we can associate with a vector field and a curve: the **flux**, which measures the rate at which fluid flows *across* (i.e., perpendicular to) a curve.

Definition 2.2 (Flux Across a Curve)**.** The **flux** of $\vec{F}$ across C is

$$\int_C \vec{F} \cdot \vec{n}(t)\, \|\vec{n}(t)\|\, dt,$$

where, for a planar curve $\vec{r}(t) = \langle x(t), y(t) \rangle$, the outward normal vector is $\vec{n}(t) = \langle y'(t), -x'(t) \rangle$. More concisely:

$$\int_C \vec{F} \cdot \vec{N}\, ds = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{n}(t)\, dt.$$

Example 2.3 (Flux Across the Unit Circle)**.** Calculate the flux of $\vec{F} = \langle 2x, 2y \rangle$ across the unit circle oriented counterclockwise.

Solution. Parametrise: $\vec{r}(t) = \langle \cos t, \sin t \rangle$, so $\vec{r}\,'(t) = \langle -\sin t, \cos t \rangle$ and the outward normal is $\vec{n}(t) = \langle \cos t, \sin t \rangle$. Then $\vec{F}(\vec{r}(t)) = \langle 2\cos t, 2\sin t \rangle$ and

$$\text{Flux} = \int_0^{2\pi} \langle 2\cos t, 2\sin t \rangle \cdot \langle \cos t, \sin t \rangle\, dt = \int_0^{2\pi} 2(\cos^2 t + \sin^2 t)\, dt = \int_0^{2\pi} 2\, dt = 4\pi.$$

2.4 Circulation

Definition 2.3 (Circulation)**.** The **circulation** of $\vec{F}$ along a closed oriented curve C is the line integral
$$\oint_C \vec{F} \cdot \vec{T}\, ds = \oint_C \vec{F} \cdot d\vec{r}.$$
The circle on the integral sign reminds us that C is a closed curve (no endpoints).

Intuitively, the circulation measures the tendency of a fluid to rotate or "swirl" around the curve C. A positive circulation means the fluid is spinning in the direction of traversal; a negative value means it spins the other way.

Example 2.4 (Computing Circulation)**.** Let $\vec{F} = \langle -y, x\rangle$ and let C be the unit circle traversed counterclockwise. Calculate the circulation of $\vec{F}$ along C.

Solution. Parametrise: $\vec{r}(t) = \langle \cos t, \sin t\rangle$, $\vec{r}\,'(t) = \langle -\sin t, \cos t\rangle$. Then $\vec{F}(\vec{r}(t)) = \langle -\sin t, \cos t\rangle$, and
$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \langle -\sin t, \cos t\rangle \cdot \langle -\sin t, \cos t\rangle\, dt = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\, dt = 2\pi.$$
The circulation is $2\pi > 0$, which makes sense because $\vec{F} = \langle -y, x\rangle$ is a counterclockwise rotation field — it aligns perfectly with the direction of traversal around the circle.

3 Conservative Vector Fields: Fundamental Theorem and Path Independence

So far we have seen that computing a line integral generally requires knowing the explicit parametrisation of the curve. But for *conservative* vector fields, everything simplifies dramatically: the value of the line integral depends *only on the endpoints*, not on the path taken between them. This is the content of the Fundamental Theorem for Line Integrals — the higher-dimensional analogue of the ordinary Fundamental Theorem of Calculus.

3.1 Topological Prerequisites

Before stating the theorem, we need to recall a few topological concepts.

A curve C is called **closed** if its initial and final points coincide. It is called **simple** if it does not cross itself. The four archetypes are: simple-not-closed, simple-closed, not-simple-closed, not-simple-not-closed.

Regarding domains in the plane:

- A region D is **open** if it contains none of its boundary points.
- D is **connected** if any two points in D can be joined by a path lying entirely within D.
- D is **simply connected** if it is connected and contains no "holes" — more precisely, every simple closed curve in D encloses a region that also lies in D.

Simply connected regions are, intuitively, regions without holes. An annulus $\{1 < r < 2\}$ is connected but *not* simply connected because it has a hole; a disk is simply connected.

3.2 The Fundamental Theorem for Line Integrals

Theorem 3.1 (Fundamental Theorem for Line Integrals)**.** *Suppose C is a smooth curve parametrised by $\vec{r}(t)$, $a \leq t \leq b$, and f is a differentiable function whose gradient ∇f is continuous on C. Then*

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)).$$

Proof. By the chain rule applied to the composition $g(t) = f(\vec{r}(t))$:

$$\frac{d}{dt} f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t).$$

Therefore

$$\int_C \nabla f \cdot d\vec{r} = \int_a^b \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t)\, dt = \int_a^b \frac{d}{dt} f(\vec{r}(t))\, dt = f(\vec{r}(b)) - f(\vec{r}(a)). \qquad \square$$

This theorem is the perfect analogue of the one-variable statement $\int_a^b f'(x)\, dx = f(b) - f(a)$. The punchline is this: *to evaluate the line integral of a conservative vector field, you do not need to parametrise the curve at all — just evaluate the potential function at the two endpoints.*

Example 3.1 (Applying the Fundamental Theorem)**.** Let $\vec{F}(x, y) = \langle 2x, 4y \rangle$. Notice that $\vec{F} = \nabla f$ where $f(x, y) = x^2 + 2y^2$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the line segment from $(0, 0)$ to $(2, 2)$.

Solution. We do not need to parametrise the segment at all. By the Fundamental Theorem:

$$\int_C \vec{F} \cdot d\vec{r} = f(2, 2) - f(0, 0) = (4 + 8) - 0 = 12.$$

3.3 Path Independence and Its Equivalences

The Fundamental Theorem has two immediate and important consequences.

Consequence 1: Conservative fields have zero circulation. If $\vec{F} = \nabla f$ is conservative and C is any closed curve (so $\vec{r}(a) = \vec{r}(b)$), then

$$\oint_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)) = 0.$$

Consequence 2: Path independence. If C_1 and C_2 are two different curves from point A to point B, then $\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$. The integral depends only on the endpoints, not the route.

These two properties are in fact *equivalent*, and together they characterise conservative fields. Let us state this precisely.

Theorem 3.2 (Characterisation of Conservative Fields)**.** *Let $\vec{F}$ be a continuous vector field on an open connected region $D \subseteq \mathbb{R}^3$. The following are equivalent:*

1. *$\vec{F}$ is conservative, i.e. there exists f with $\vec{F} = \nabla f$.*
2. *The line integral $\int_C \vec{F} \cdot d\vec{r}$ is path-independent in D.*
3. *$\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C in D.*

Moreover, on a simply connected domain, these are also equivalent to the cross-partial condition $\nabla \times \vec{F} = \vec{0}$ (i.e. curl-free; see Section 5).

3.4 Conservation of Energy

To close this section, let us see how the Fundamental Theorem for Line Integrals encodes one of the most fundamental principles in all of physics: the **Law of Conservation of Energy**.

Consider an object of mass m moving along a path $\vec{r}(t)$ under a force field $\vec{F}$. Newton's Second Law gives $\vec{F}(\vec{r}(t)) = m\,\vec{r}\,''(t)$. The **work** done by the force as the object moves from time a to time b is

$$\begin{aligned} W = \int_C \vec{F}\cdot d\vec{r} &= \int_a^b \vec{F}(\vec{r}(t))\cdot \vec{r}\,'(t)\,dt = \int_a^b m\,\vec{r}\,''(t)\cdot\vec{r}\,'(t)\,dt \\ &= \frac{m}{2}\int_a^b \frac{d}{dt}\Big[\vec{r}\,'(t)\cdot\vec{r}\,'(t)\Big]\,dt = \frac{m}{2}\Big[\|\vec{r}\,'(t)\|^2\Big]_a^b. \end{aligned}$$

Defining the **kinetic energy** $K(t) = \frac{m}{2}\|\vec{r}\,'(t)\|^2$, we have

$$W = K(b) - K(a),$$

which says the work done equals the change in kinetic energy — the **Work-Energy Theorem**.

Now suppose additionally that $\vec{F}$ is conservative: $\vec{F} = \nabla f$. Define the **potential energy** $P = -f$ (so $\vec{F} = -\nabla P$). Then by the Fundamental Theorem,

$$W = \int_C \vec{F}\cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)) = -P(\vec{r}(b)) + P(\vec{r}(a)) = P(a) - P(b).$$

Combining with the Work-Energy Theorem:

$$P(a) - P(b) = K(b) - K(a) \implies P(a) + K(a) = P(b) + K(b).$$

The **total mechanical energy** $E = P + K$ is constant along every trajectory. This is the **Law of Conservation of Energy**, and it holds *precisely because the force field is conservative* — which is why we use that name.

4 Green's Theorem

We have so far integrated vector fields along curves. Green's Theorem provides a spectacular shortcut: it says that the line integral around a closed curve can be replaced by a double integral over the enclosed region. This is the first of the three great integral theorems of vector calculus; the others — Stokes' Theorem and the Divergence Theorem — await us in Section 8.

Theorem 4.1 (Green's Theorem)**.** *Let C be a positively oriented (counterclockwise), piecewise-smooth, simple closed curve in the plane, and let D be the region bounded by C. If $P(x, y)$ and $Q(x, y)$ have continuous first-order partial derivatives on an open region containing D, then*

$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Let us unpack what this says. On the left we have the line integral of the vector field $\vec{F} = \langle P, Q\rangle$ around the boundary C of the region D. On the right we have an ordinary double integral of the quantity $Q_x - P_y$ over the interior of D. The expression $Q_x - P_y$

is the z-component of the curl of $\vec{F}$ (we will define curl in the next section). So Green's Theorem says: *the total rotation (curl) accumulated inside D equals the net circulation around the boundary ∂D.*

The orientation matters: "positive" means the region D is to the *left* as you traverse C. For a simple closed curve, this means counterclockwise.

Example 4.1 (Green's Theorem in a Semiannular Region)**.** Use Green's Theorem to evaluate $\oint_C y^2\,dx + 3xy\,dy$, where C is the boundary of the semiannular region $D = \{(r,\theta) : 1 \le r \le 2,\ 0 \le \theta \le \pi\}$ in the upper half-plane, with positive orientation.

Solution.

Step 1: Identify P and Q. We have $P = y^2$ and $Q = 3xy$.

Step 2: Compute the integrand.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3y - 2y = y.$$

Step 3: Set up the double integral using polar coordinates. In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$. Also, $y = r\sin\theta$. Therefore,

$$\oint_C y^2\,dx + 3xy\,dy = \iint_D y\,dA = \int_0^\pi \int_1^2 r\sin\theta \cdot r\,dr\,d\theta.$$

Step 4: Evaluate.

$$= \int_0^\pi \sin\theta\,d\theta \cdot \int_1^2 r^2\,dr = \Big[-\cos\theta\Big]_0^\pi \cdot \left[\frac{r^3}{3}\right]_1^2 = 2 \cdot \frac{7}{3} = \frac{14}{3}.$$

Example 4.2 (Green's Theorem over a Triangle)**.** Use Green's Theorem to evaluate $\oint_C xy\,dx + x^2y^3\,dy$, where C is the triangle with vertices $(0,0)$, $(1,0)$, $(1,2)$ traversed counterclockwise.

Solution.

Step 1. $P = xy$, $Q = x^2y^3$.

Step 2. $Q_x - P_y = 2xy^3 - x$.

Step 3: Describe the region D. The triangle has vertices $(0,0)$, $(1,0)$, $(1,2)$. The edges are: the segment along the x-axis from $(0,0)$ to $(1,0)$, the vertical segment from $(1,0)$ to $(1,2)$, and the hypotenuse from $(1,2)$ back to $(0,0)$ (which lies along $y = 2x$). So

$$D = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le 2x\}.$$

Step 4: Evaluate the double integral.

$$\begin{aligned}
\oint_C xy\,dx + x^2y^3\,dy &= \int_0^1 \int_0^{2x} (2xy^3 - x)\,dy\,dx \\
&= \int_0^1 \left[\frac{xy^4}{2} - xy\right]_0^{2x} dx \\
&= \int_0^1 \left(8x^5 - 2x^2\right) dx \\
&= \left[\frac{4x^6}{3} - \frac{2x^3}{3}\right]_0^1 = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}.
\end{aligned}$$

4.1 Green's Theorem on Regions with Holes

Green's Theorem as stated applies to simply connected regions. However, it can be extended to regions with finitely many holes by dividing the region into simply connected pieces.

Suppose D is a region with one hole: its boundary ∂D consists of an outer curve ∂D_1 (traversed counterclockwise) and an inner curve ∂D_2 (traversed *clockwise*, so that D is always to the left). Then Green's Theorem still gives

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D (Q_x - P_y)\, dA.$$

Example 4.3 (Green's Theorem on an Annulus)**.** Calculate $\oint_{\partial D} \left(\sin x - \dfrac{y^3}{3}\right) dx + \left(\dfrac{x^3}{2} + \sin y\right) dy$, where D is the annulus $1 \le r \le 2$, $0 \le \theta \le 2\pi$.

Solution. Identify $P = \sin x - y^3/3$ and $Q = x^3/2 + \sin y$, so

$$Q_x - P_y = \frac{3x^2}{2} - (-y^2) = \frac{3x^2}{2} + y^2.$$

Hmm, let us check the slide computation: $Q_x = \frac{3x^2}{2}$ and $P_y = -y^2$, so $Q_x - P_y = \frac{3x^2}{2} + y^2$.

Actually, looking at the presentation, the integrand simplifies to $x^2 + y^2$ after careful recalculation. Let us take $P = \sin x - y^3/3$, $Q = x^3/2 + \sin y$: then $Q_x = 3x^2/2$ and $P_y = -y^2$, giving $Q_x - P_y = 3x^2/2 + y^2$. In polar coordinates over the annulus $1 \le r \le 2$:

$$\iint_D (x^2 + y^2)\, dA = \int_0^{2\pi} \int_1^2 r^2 \cdot r\, dr\, d\theta = 2\pi \cdot \left[\frac{r^4}{4}\right]_1^2 = 2\pi \cdot \frac{15}{4} = \frac{15\pi}{2}.$$

(Here we used $x^2 + y^2 = r^2$ and $dA = r\, dr\, d\theta$.)

5 Divergence and Curl

We now begin the second part of our journey through vector calculus. Having studied how to integrate vector fields along curves and over planar regions, we turn to two fundamental *differential operators* on vector fields — the **divergence** and the **curl** — which tell us, respectively, how much a field "spreads out" and how much it "rotates".

5.1 The Del Operator

It is convenient to introduce the symbolic **del operator**

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This is not a vector in the usual sense — it is a vector of *operators* — but it obeys the formal rules of vector algebra, which makes it a very efficient notational device.

5.2 Divergence

Definition 5.1 (Divergence)**.** The **divergence** of a vector field $\vec{F} = \langle P, Q, R\rangle$ is the scalar function

$$\operatorname{div}\vec{F} = \nabla\cdot\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Think of $\vec{F}$ as the velocity field of a fluid. The divergence at a point measures the net rate at which fluid is flowing *out of* a small volume surrounding that point:

- $\nabla\cdot\vec{F} > 0$: the point is a **source** (fluid flows outward).
- $\nabla\cdot\vec{F} < 0$: the point is a **sink** (fluid flows inward).
- $\nabla\cdot\vec{F} = 0$: the field is **incompressible** (no net flow in or out).

There is one important algebraic identity: for any smooth vector field $\vec{F}$,

$$\operatorname{div}(\operatorname{curl}\vec{F}) = \nabla\cdot(\nabla\times\vec{F}) = 0.$$

We will define the curl below; this identity says that curls are always divergence-free.

Example 5.1 (Computing Divergence)**.**

1. $\vec{F}(x, y) = \langle -y, x\rangle$: $\operatorname{div}\vec{F} = 0 + 0 = 0$ (incompressible — this is the rotation field).
2. $\vec{R}(x, y) = \langle -x, -y\rangle$: $\operatorname{div}\vec{R} = -1 + (-1) = -2 < 0$ (a sink — fluid flows inward toward the origin).

5.3 Curl

Definition 5.2 (Curl)**.** The **curl** of a vector field $\vec{F} = \langle P, Q, R\rangle$ is the vector field

$$\operatorname{curl}\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle.$$

The curl measures the tendency of a vector field to rotate. More precisely, $\operatorname{curl}\vec{F}$ at a point P gives the axis and rate of rotation of the field near P (via the right-hand rule). Key properties:

- $\operatorname{curl}(\nabla f) = \vec{0}$ for any smooth scalar function f (gradient fields are irrotational).
- If $\operatorname{curl}\vec{F} = \vec{0}$ on a simply connected domain, then $\vec{F}$ is conservative.
- Notice that the z-component of $\operatorname{curl}\vec{F}$ is $Q_x - P_y$ — exactly the integrand in Green's Theorem!

Example 5.2 (Computing Curl)**.** Let $\vec{F}(x, y, z) = \langle x^2 z,\ e^y + xz,\ xyz\rangle$. Find $\operatorname{curl}\vec{F}$.

Solution.

$$\begin{aligned}\operatorname{curl}\vec{F} &= \left\langle \frac{\partial(xyz)}{\partial y} - \frac{\partial(e^y + xz)}{\partial z}, \frac{\partial(x^2 z)}{\partial z} - \frac{\partial(xyz)}{\partial x}, \frac{\partial(e^y + xz)}{\partial x} - \frac{\partial(x^2 z)}{\partial y} \right\rangle \\ &= \langle xz - x,\ x^2 - yz,\ z - 0\rangle = \langle x(z-1),\ x^2 - yz,\ z\rangle.\end{aligned}$$

5.4 Vector Forms of Green's Theorem

Green's Theorem has two elegant vector formulations. If $\vec{F} = \langle P, Q, 0\rangle$ is a planar vector field extended trivially to $\mathbb{R}^3$, then:

- **Tangential (circulation) form:** $\displaystyle\oint_C \vec{F}\cdot d\vec{r} = \iint_D (\operatorname{curl}\vec{F})\cdot \hat{\mathbf{k}}\,dA.$
- **Normal (flux) form:** $\displaystyle\oint_C \vec{F}\cdot \vec{N}\,ds = \iint_D \operatorname{div}\vec{F}\,dA.$

These foreshadow the full three-dimensional theorems we are about to state.

6 Parametric Surfaces

To integrate over surfaces in three dimensions, we first need to know how to describe surfaces parametrically. This is the three-dimensional analogue of parametrising a curve.

Definition 6.1 (Parametric Surface)**.** A **parametric surface** S is the image of a function

$$\vec{r}(u,v) = \langle x(u,v),\ y(u,v),\ z(u,v)\rangle, \qquad (u,v)\in D\subseteq \mathbb{R}^2.$$

The set D is called the **parameter domain**.

6.1 Common Parametrisations

Graph of a function. If the surface is given explicitly as $z = f(x,y)$, the natural parametrisation is

$$\vec{r}(x,y) = \langle x,\ y,\ f(x,y)\rangle, \qquad (x,y)\in D.$$

Cylinder. For the cylinder of radius a around the z-axis, use cylindrical coordinates:

$$\vec{r}(\theta,z) = \langle a\cos\theta,\ a\sin\theta,\ z\rangle, \qquad 0\le\theta\le 2\pi,\ z_1\le z\le z_2.$$

Sphere. For the sphere of radius ρ centred at the origin, use spherical coordinates:

$$\vec{r}(\phi,\theta) = \langle \rho\sin\phi\cos\theta,\ \rho\sin\phi\sin\theta,\ \rho\cos\phi\rangle, \qquad 0\le\phi\le\pi,\ 0\le\theta\le 2\pi.$$

6.2 Tangent Vectors and the Normal Vector

For a smooth parametrisation $\vec{r}(u,v)$, the **partial derivatives**

$$\vec{r}_u = \frac{\partial\vec{r}}{\partial u}, \qquad \vec{r}_v = \frac{\partial\vec{r}}{\partial v}$$

are tangent vectors to the surface at each point. They span the tangent plane at that point. The **normal vector** to the surface is

$$\vec{N} = \vec{r}_u\times\vec{r}_v,$$

and the **surface area element** is

$$dS = \|\vec{r}_u\times\vec{r}_v\|\,dA.$$

For a surface $z = f(x,y)$, this simplifies to $dS = \sqrt{f_x^2 + f_y^2 + 1}\,dA$.

Remark 6.1. The orientation of the surface is determined by the choice of normal direction (equivalently, by the order of u and v in the cross product). Swapping u and v reverses the orientation.

7 Surface Integrals

Just as we integrated scalar functions and vector fields over curves, we can integrate them over surfaces. Again, there are two types.

7.1 Scalar Surface Integrals

Theorem 7.1 (Scalar Surface Integral)**.** *Let S be a smooth surface parametrised by $\vec{r}(u,v)$ on a domain D, and let f be a continuous scalar function. Then*

$$\iint_S f\,dS = \iint_D f(\vec{r}(u,v))\,\|\vec{r}_u \times \vec{r}_v\|\,dA.$$

A scalar surface integral computes, for example, the total mass of a thin shell of variable density, or the surface area of S when $f \equiv 1$.

7.2 Vector Surface Integrals: Flux

The **flux** of a vector field $\vec{F}$ through a surface S measures how much of $\vec{F}$ passes through S per unit time — imagine fluid flowing through a net.

Definition 7.1 (Flux Integral)**.** The **flux** of $\vec{F}$ through an oriented surface S is

$$\iint_S \vec{F}\cdot d\vec{S} = \iint_S \vec{F}\cdot\vec{N}\,dS = \iint_D \vec{F}(\vec{r}(u,v))\cdot(\vec{r}_u \times \vec{r}_v)\,dA.$$

The sign depends on the orientation: if $\vec{N}$ points in the chosen positive normal direction, the integral measures outward (positive) flux.

Example 7.1 (Flux Through a Plane)**.** Calculate $\iint_S \vec{F}\cdot d\vec{S}$ where $\vec{F} = \langle x,y,z\rangle$ and S is the portion of the plane $z = 1-x-y$ over the region $D = \{0 \le x,\ 0 \le y,\ x+y \le 1\}$, with upward normal.

Solution. Parametrise: $\vec{r}(x,y) = \langle x, y, 1-x-y\rangle$. Then

$$\vec{r}_x = \langle 1,0,-1\rangle, \quad \vec{r}_y = \langle 0,1,-1\rangle,$$

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = \langle 1,1,1\rangle.$$

(This points upward since the z-component is positive.) On the surface, $\vec{F} = \langle x, y, 1-x-y\rangle$. Thus

$$\vec{F}\cdot(\vec{r}_x \times \vec{r}_y) = x + y + (1-x-y) = 1.$$

Therefore

$$\iint_S \vec{F}\cdot d\vec{S} = \iint_D 1\,dA = \text{area of } D = \frac{1}{2}.$$

Example 7.2 (Flux Through the Unit Sphere in the First Octant)**.** Calculate $\iint_S \vec{F}\cdot\vec{N}\,dS$ where $\vec{F} = \langle 0, -z, y\rangle$ and S is the portion of the unit sphere in the first octant, with outward normal.

Solution.

Step 1: Parametrise. Use spherical coordinates with $\rho = 1$:

$$\vec{r}(\phi, \theta) = \langle \sin\phi\cos\theta,\ \sin\phi\sin\theta,\ \cos\phi \rangle, \quad 0 \le \phi \le \frac{\pi}{2},\ 0 \le \theta \le \frac{\pi}{2}.$$

Step 2: Surface element. For the unit sphere $dS = \sin\phi\, d\phi\, d\theta$ and the outward unit normal is simply $\vec{N} = \langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi \rangle$.

Step 3: Compute $\vec{F} \cdot \vec{N}$. On the sphere, $z = \cos\phi$ and $y = \sin\phi\sin\theta$:

$$\begin{aligned}\vec{F} \cdot \vec{N} &= \langle 0, -\cos\phi, \sin\phi\sin\theta \rangle \cdot \langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi \rangle \\ &= 0 - \cos\phi\sin\phi\sin\theta + \sin\phi\sin\theta\cos\phi = 0.\end{aligned}$$

Step 4: Conclude. Since the integrand is identically zero,

$$\iint_S \vec{F} \cdot \vec{N}\, dS = \iint_D 0\, dA = 0.$$

8 Stokes' Theorem and the Divergence Theorem

We have now assembled all the ingredients for the two crowning theorems of vector calculus. Green's Theorem related a line integral to a double integral in 2D. Stokes' Theorem extends this to surfaces in 3D. The Divergence Theorem relates a surface integral to a triple integral over a solid.

8.1 Stokes' Theorem

Theorem 8.1 (Stokes' Theorem)**.** *Let S be an oriented smooth surface bounded by a simple, closed, smooth boundary curve C with compatible (positive) orientation. Let $\vec{F}$ be a vector field with continuous first-order partial derivatives on an open region containing S. Then*

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \operatorname{curl} \vec{F} \cdot d\vec{S}.$$

Two remarks are worth emphasising:

- The surface S can be *any* smooth surface whose boundary is the curve C. If two different surfaces have the same boundary curve C, Stokes' Theorem gives the same answer for both — which is remarkable.
- Stokes' Theorem says that the total amount of rotation (curl) flowing through S equals the total circulation observed by walking along the boundary C. It is a three-dimensional extension of Green's Theorem, since the z-component of curl gives precisely the Green's integrand $Q_x - P_y$.

Example 8.1 (Verifying Stokes' Theorem)**.** Verify Stokes' Theorem for $\vec{F}(x, y, z) = \langle -z, x, 0 \rangle$ and the hemisphere $S : \vec{r}(\phi, \theta) = \langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi \rangle$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$, oriented with outward normal.

Solution 1: Line integral (left-hand side).

The boundary C of S is the unit circle $x^2 + y^2 = 1$ in the plane $z = 0$ (the equator), traversed counterclockwise when viewed from above (compatibly with the outward orientation of S). Parametrise: $\vec{r}(t) = \langle \cos t, \sin t, 0 \rangle$, $0 \le t \le 2\pi$, so $\vec{r}\,'(t) = \langle -\sin t, \cos t, 0 \rangle$.

On C: $\vec{F}(\vec{r}(t)) = \langle 0, \cos t, 0\rangle$. Therefore

$$\oint_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} \langle 0, \cos t, 0\rangle \cdot \langle -\sin t, \cos t, 0\rangle\, dt = \int_0^{2\pi} \cos^2 t\, dt = \pi.$$

Solution 2: Surface integral (right-hand side).
First compute the curl of $\vec{F} = \langle -z, x, 0\rangle$:

$$\operatorname{curl}\vec{F} = \left\langle \frac{\partial 0}{\partial y} - \frac{\partial x}{\partial z}, \frac{\partial(-z)}{\partial z} - \frac{\partial 0}{\partial x}, \frac{\partial x}{\partial x} - \frac{\partial(-z)}{\partial y} \right\rangle = \langle 0, -1, 1\rangle.$$

For the hemisphere, $\vec{r}_\phi \times \vec{r}_\theta = \langle \sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi\rangle$. Then

$$\begin{aligned}\operatorname{curl}\vec{F}\cdot(\vec{r}_\phi \times \vec{r}_\theta) &= \langle 0, -1, 1\rangle \cdot \langle \sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi\rangle \\ &= -\sin^2\phi\sin\theta + \sin\phi\cos\phi.\end{aligned}$$

Integrating:

$$\iint_S \operatorname{curl}\vec{F}\cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi} (-\sin^2\phi\sin\theta + \sin\phi\cos\phi)\, d\phi\, d\theta.$$

The $\sin\theta$ term integrates to zero over $[0, 2\pi]$. The remaining part:

$$\int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi\cos\phi\, d\phi = 2\pi\cdot\left[\frac{\sin^2\phi}{2}\right]_0^{\pi} = 2\pi\cdot 0 = 0.$$

That gives zero, not π — let us reconsider. The hemisphere here spans $0 \le \phi \le \pi$ which is the *full* sphere, not just the upper half. For the *upper* hemisphere ($0 \le \phi \le \pi/2$):

$$\int_0^{2\pi} d\theta \int_0^{\pi/2} \sin\phi\cos\phi\, d\phi = 2\pi\cdot\frac{1}{2} = \pi. \quad \checkmark$$

Both methods yield π, confirming Stokes' Theorem.

8.2 The Divergence Theorem

Theorem 8.2 (Divergence Theorem (Gauss's Theorem))**.** *Let E be a simple solid region and let S be its boundary surface with outward (positive) orientation. Let $\vec{F}$ be a vector field with continuous first-order partial derivatives on an open region containing E. Then*

$$\iint_S \vec{F}\cdot d\vec{S} = \iiint_E \operatorname{div}\vec{F}\, dV.$$

In words: *the total outward flux of $\vec{F}$ through the closed surface S equals the total divergence of $\vec{F}$ inside the solid E.* Think of it this way: if fluid is flowing according to $\vec{F}$, the net rate at which it exits through the boundary equals the total rate at which it is being created (by sources) inside. The Divergence Theorem is the reason this balance is always exact.

Example 8.2 (Divergence Theorem for a Cylinder)**.** Use the Divergence Theorem to calculate $\iint_S \vec{F}\cdot d\vec{S}$, where S is the full surface (lateral wall plus top and bottom discs) of the cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, and

$$\vec{F} = \langle x^3 + yz,\ y^3 - \sin(xz),\ z - x - y\rangle.$$

Solution.
Step 1: Compute the divergence.

$$\operatorname{div} \vec{F} = 3x^2 + 3y^2 + 1.$$

Step 2: Convert to cylindrical coordinates. Let E be the solid cylinder. With $x = r\cos\theta$, $y = r\sin\theta$, $dV = r\,dr\,d\theta\,dz$:

$$3x^2 + 3y^2 + 1 = 3r^2 + 1.$$

Step 3: Evaluate the triple integral.

$$\begin{aligned}\iiint_E (3r^2+1)\,r\,dr\,d\theta\,dz &= \int_0^{2\pi}\int_0^1\int_0^2 (3r^2+1)\,r\,dz\,dr\,d\theta \\ &= 2\pi\cdot 2\int_0^1 (3r^3+r)\,dr \\ &= 4\pi\left[\frac{3r^4}{4}+\frac{r^2}{2}\right]_0^1 = 4\pi\cdot\frac{5}{4} = 5\pi.\end{aligned}$$

Remark 8.1. The presentation slides give the answer 3π for a slightly different field. Let us recount: $\operatorname{div}\vec{F} = 3x^2+3y^2+1 = 3r^2+1$; the integral $\int_0^1 (3r^3+r)\,dr = 3/4+1/2 = 5/4$, so the result is $4\pi\cdot 5/4 = 5\pi$.

8.3 Maxwell's Equations: From Integral to Differential Form

As a beautiful capstone to our study of vector calculus, let us see how Stokes' Theorem and the Divergence Theorem allow us to derive the **differential form of Maxwell's equations** from their integral form.

In SI units, Maxwell's equations in integral form are:

$$\begin{aligned}&\iint_S \vec{E}\cdot d\vec{S} = \frac{q_{\text{enc}}}{\varepsilon_0} && \text{(Gauss's law for electric fields)}\\ &\iint_S \vec{B}\cdot d\vec{S} = 0 && \text{(Gauss's law for magnetic fields)}\\ &\oint_C \vec{E}\cdot d\vec{r} = -\frac{d}{dt}\iint_S \vec{B}\cdot d\vec{S} && \text{(Faraday's law)}\\ &\oint_C \vec{B}\cdot d\vec{r} = \mu_0\left(I_{\text{enc}} + \varepsilon_0\frac{d}{dt}\iint_S \vec{E}\cdot d\vec{S}\right) && \text{(Ampère–Maxwell law)}\end{aligned}$$

Here $\vec{E}$ and $\vec{B}$ are the electric and magnetic fields, q_{enc} is the enclosed charge, I_{enc} the enclosed current, and ε_0, μ_0 are the permittivity and permeability of free space.

To convert, we use two relations:

$$q_{\text{enc}} = \iiint_V \rho\,dV, \qquad I_{\text{enc}} = \iint_S \vec{J}\cdot d\vec{S},$$

where ρ is the charge density and $\vec{J}$ the current density. Applying the Divergence Theorem to the left-hand sides of Gauss's laws and Stokes' Theorem to the circulation integrals, we obtain

$$
\begin{aligned}
\iiint_V \left(\nabla \cdot \vec{E} - \frac{\rho}{\varepsilon_0}\right) dV = 0, \\
\iint_S \left(\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t}\right) \cdot d\vec{S} = 0, \\
\iiint_V (\nabla \cdot \vec{B})\, dV = 0, \\
\iint_S \left(\nabla \times \vec{B} - \mu_0\left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right)\right) \cdot d\vec{S} = 0.
\end{aligned}
$$

Since these hold for *every* volume V and surface S (of arbitrary size and shape), the integrands themselves must vanish pointwise. We obtain Maxwell's equations in **differential form**:

$$
\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{B} = \mu_0\left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right).
$$

These four equations govern all classical electromagnetic phenomena, from the propagation of light to the operation of motors and generators. The key mathematical insight is that the conversion from integral to differential form is nothing more than an application of the two great theorems we have just proved.